Health Wearable Device

PCB Design — Component Analysis Report

Band01_Rev1 | Project Work — Task 1

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52	3	BLE 5.1	34×40mm
Components	Sensors	Wireless	PCB Size

What Is This Device?

Before looking at components — understanding the purpose

Purpose & Context

- A wrist-worn smartband that continuously monitors patient health
- Collects 3 vital signs: heart rate, blood oxygen (SpO2), and body temperature
- Also tracks motion and activity via accelerometer
- Transmits all data wirelessly to a smartphone or hospital gateway
- Designed for remote patient monitoring — no hospital visit needed
- Designed by Ricardo K. and Gabriel P. — PCB manufactured by Aisler

Why is wearable design challenging?

⚠ Very small PCB (34×40mm) — every mm^2 matters

⚠ Must run on a tiny battery for 73+ hours

⚠ Sensors need precision power — noise ruins readings

⚠ BLE radio must coexist with analogue sensor circuits

Key Design Specs

PCB size:	34×40mm
Wireless:	BLE 5.1 @ 2.4GHz
Power:	Battery → 3.3V + 1.8V
Sensors:	3 ICs on one I^2C bus

Teaching note — Why study the PCB before the firmware?

Every firmware decision (which I^2C address to use, which DIO pin to read, how often to poll a sensor) maps directly to a physical component on this PCB. Understanding the hardware first means the firmware code will make sense immediately when you study it in Task 2.

System Overview — Block Diagram

Band01_Rev1 — all major subsystems and how they connect

MAX30102
HR / SpO2

MAX30205
Temperature

LIS331DL
Accelerometer

I^2C Bus
DIO_5/DIO_6

CC2640R2F
ARM Cortex-M3 | BLE 5.1
128kB Flash | 28kB SRAM

LP5907 U101
3.3V LDO

LP5907 U102
1.8V LDO

PCB Antenna
BLE 2.4GHz IFA

VBAT → U101 (3.3V) + U102 (1.8V) | Y1 24MHz + Y2 32kHz crystals | FL1 ferrite bead | P1 18-pin JTAG header | L1 10µH DC-DC inductor | Passives: 27 capacitors + 11 resistors

Teaching note — I^2C bus topology

All 3 sensors share exactly 2 wires (SCL clock + SDA data). The CC2640R2F is the bus master — it initiates every transaction. Each sensor has a unique 7-bit address (0x57, 0x90, 0x19) so the master can talk to one at a time. Pull-up resistors R103/R104 (10kΩ) keep the lines HIGH when idle.

Microcontroller — CC2640R2FRSMR (U1)

Texas Instruments | 32-pin VFQFN (RSM) | The brain of the entire wearable system

Hardware Specifications

Architecture:	ARM Cortex-M3 @ up to 48MHz
Flash:	128 kB (stores the firmware program)
SRAM:	28 kB (runtime data & stack)
GPIO:	10 pins: DIO_0 – DIO_9
Interfaces:	I^{2}C, SPI, UART, ADC
Supply voltage:	1.8V – 3.8V (powered from 3.3V rail)

Key Pin Assignments

DIO_5 / DIO_6	I^{2}C SCL / SDA — shared by all 3 sensors
DIO_0 / DIO_1	Push buttons SW101 / SW102 user input
DIO_8	MAX30102 interrupt — signals new HR data ready
DIO_9 / DIO_2	LIS331 motion interrupts INT_1 / INT_2
DIO_3 / DIO_4	JTAG TCK / TMS — Code Composer Studio
DCDC_SW (18)	Internal DC-DC → L1 (10µH) → VDDS rail
RF_P / RF_N	BLE RF output → L12/L21/C14 → antenna

What is ARM Cortex-M3?

ARM is a processor architecture (like x86 in your PC but much smaller and lower power). Cortex-M3 is a 32-bit core optimised for microcontrollers — it runs real firmware code, handles interrupts, and manages peripherals. TI adds the BLE radio, memory, and I/O pins around this core.

What is JTAG?

JTAG (Joint Test Action Group) is a standard 4-wire debug interface. It lets a PC tool (Code Composer Studio) pause the chip mid-execution, inspect memory, set breakpoints, and flash new firmware. The XDS110 probe converts USB on the PC side to JTAG signals on the chip side.

Teaching note — Why does flash size matter in embedded systems?

Unlike a PC that loads programs from an SSD, an embedded MCU stores firmware permanently in its own flash. 128kB must hold: the BLE stack (~50kB), all application code, font bitmaps, and RTOS. This is why every byte of code matters in embedded development — there is no 'more storage'.

Sensor ICs — How Vital Signs Are Measured

Three sensors, one I²C bus — measuring heart rate, blood oxygen, temperature, and motion

MAX30102EFD+ — Heart Rate & SpO2 | Analog Devices | 14-pin PSON | I²C 0x57 | 1.8V supply

How PPG works: The sensor shines a red LED (660nm) and IR LED (880nm) through the skin. A photodetector measures the light that bounces back. With each heartbeat, blood vessels expand slightly — more blood absorbs more light. The chip counts these pulses to calculate BPM. For SpO2: oxygenated haemoglobin absorbs IR more than red; deoxygenated absorbs red more. The ratio of red/IR absorption gives the SpO2% (normal = 95–100%). LED current set to 6.2mA in firmware. Interrupt on DIO_8 fires when new sample is ready.

MAX30205MTA+T — Body Temperature | Analog Devices | 9-pin TDFN 3×3 | I²C 0x90 | 3.3V supply

Clinical accuracy ±0.1°C with 16-bit resolution (0.00390625°C per bit). Range: 0°C–45°C — covers all human body temperatures including fever. Internally uses a silicon bandgap temperature sensor — the forward voltage of a p-n junction changes predictably with temperature. I²C address 0x90 (address pins A0/A1/A2 = GND). OS (overtemperature shutdown) pin can fire a hardware interrupt if temperature exceeds a threshold set in firmware — enabling automatic fever alerts without CPU polling.

LIS331DLHTR — 3-Axis Accelerometer | STMicroelectronics | 16-pin LGA | I²C 0x19 | 3.3V supply

MEMS (Micro-Electro-Mechanical Systems) capacitive accelerometer — microscopic silicon proof masses suspended by springs detect acceleration by measuring capacitance change. Measures X/Y/Z acceleration at ±2g/±4g/±8g (selectable). Data rate: 50Hz–1kHz configurable. Uses: step counting, fall detection, posture analysis, motion artifact removal from PPG signal. Two interrupt pins (INT_1 → DIO_9, INT_2 → DIO_2) for autonomous event detection — free-fall, threshold crossing, orientation change — without waking the CPU.

Teaching note — Why use interrupts instead of polling?

Polling means the CPU constantly asks 'do you have data yet?' in a loop — wasting power. Interrupts let the sensor say 'I have data NOW' via a dedicated pin (DIO_8 for MAX30102). The CPU sleeps between measurements, achieving the <1µA deep sleep current critical for 73-hour battery life. This interrupt-driven design is a core embedded systems principle.

Power Architecture — Dual-Rail Design

VBAT (LiPo battery) → two LP5907 LDO regulators → clean 3.3V and 1.8V rails with firmware-controlled power gating

VBAT

LiPo ~3.7V

U101 — LP5907MFX-3.3

Output: 3.3V (NL3V3)

U102 — LP5907MFX-1.8

Output: 1.8V (NL1V8)

3.3V consumers:

CC2640R2F digital I/O • LIS331DL accelerometer • MAX30205 temperature • Push buttons • I^{2}C pull-up resistors R103/R104

1.8V consumer:

MAX30102 only — the heart rate sensor's analogue front end needs 1.8V for accurate optical measurements

What is an LDO regulator?

LDO = Low Dropout. It converts a higher voltage (VBAT ~3.7V) to a stable lower voltage (3.3V or 1.8V) even when the input is only slightly higher than the output. 'Ultra-low noise' matters here: LP5907 noise = 6.4µVRMS — essential because the MAX30102 optical sensor is measuring tiny light intensity changes. Any noise on its power supply appears as fake heartbeat signals.

Internal DC-DC Converter + Power Gating

DCDC_SW (pin 18) → L1 10µH (Taiyo Yuden) → C5/C8 10µF → VDDS rail. The switching converter is more efficient than a pure LDO for the CPU core supply. Both LP5907 EN pins are connected to CC2640R2F GPIOs — firmware can shut off sensors completely during BLE sleep, achieving <1µA standby and ~73h on 150mAh battery.

Teaching note — Why does the MAX30102 need its own separate 1.8V rail?

The MAX30102 analogue front end (the optical sensor circuitry) is extremely sensitive to power supply noise. A shared noisy rail would contaminate the tiny photocurrent signals it measures, producing false heart rate readings. Giving it a dedicated ultra-low-noise 1.8V supply isolates it completely. This is a classic analogue/digital separation principle in mixed-signal PCB design.

RF, Timing & Support Components

Crystals, RF matching, ferrite bead, and expansion header — each has a precise engineering purpose

24MHz Crystal Y1

NX2016SA (NDK) | X24M_P/N pins 31/30

BLE requires a precise 24MHz clock for its 2.4GHz radio frequency synthesis (2400MHz = N × 24MHz). Without this crystal the BLE radio cannot operate. Load caps C22/C23 marked DNM — CC2640R2F has sufficient internal capacitance. Connects to X24M_P (pin 31) and X24M_N (pin 30).

32kHz Crystal Y2

ECS-.327-7-34B-TR | X32K_Q1/Q2 pins 5/6

The low-power real-time clock crystal. During deep sleep the 24MHz crystal is off (saves power), but Y2 keeps ticking at 32.768kHz (~1µA). It wakes the chip at precise BLE advertising intervals (every 100ms by default). Load caps C1/C2 (12pF) are populated. The 'magic' frequency 32.768kHz = 2^{15} — easy to divide into seconds.

RF Inductors L12 + L21

Murata LQP03T 0201 | 2nH + 15nH

Impedance matching network between CC2640R2F RF pins and the 50Ω PCB antenna. The CC2640R2F RF port is NOT 50Ω — L21 (15nH) + L12 (2nH) + C14 (12pF) transform it to 50Ω at 2.4GHz. Without this network, most of the BLE transmit power reflects back instead of radiating — range would be severely reduced. Designed per TI reference design.

PCB Trace Antenna

TI AN043 SWRA117D — Inverted-F (IFA)

A copper trace shaped like an upside-down F etched onto the PCB top layer. No separate component needed. Inverted-F antennas are common in wearables: compact, cheap, and omnidirectional. Geometry follows TI application note AN043. The transmission line connecting it to L12 is exactly 50Ω impedance (line width calculated per JLCPCB PCB specs).

Ferrite Bead FL1

Murata BLM18HE152SN1D | 0603

Acts as a frequency-dependent resistor: near-zero impedance at DC (power passes freely), 1500Ω impedance at 100MHz (RF noise is blocked). Placed in series on the 3.3V external supply input line. Prevents RF interference radiated by the BLE antenna from coupling backwards into the sensitive analogue sensor supply rails — a key EMI suppression technique.

Expansion Header P1

Samtec HTSW-109-07-G-D | 18-pin 9×2

18-pin through-hole header exposing every key signal: DIO_0–DIO_9, JTAG_TCK/TMS/TDI/TDO, nRESET, VBAT, 3V3, GND. Primary development interface — TI XDS110 JTAG probe connects here for CCS firmware programming and debugging. Also used in the prototype phase with a Raspberry Pi for serial sensor data readout. Essential for student development and testing.

Passive Components — Capacitors & Resistors

27 capacitors + 11 resistors — each placed for a specific electrical engineering reason

Capacitors — MLCCs (Murata, Samsung, KEMET) | 0201 and 0402 packages

Reference	Value / Package	Location	Engineering Purpose
C1, C2	12pF / 0201	32kHz crystal Y2	Load capacitors — set resonant frequency of Y2 crystal precisely
C3,C4,C6,C9,C10,C20	0.1µF / 0201	CC2640R2F supply pins	High-freq decoupling — absorbs switching noise at each supply pin
C5, C8, C106, C109	10µF / 0402	DC-DC converter + sensor rails	Bulk energy storage — handles large current transients at startup
C14	12pF / 0201	RF matching network	Shunt capacitor in LC matching — tunes antenna impedance at 2.4GHz
C19,C101,C103,C104,C107	1µF / 0402	LP5907 regulators	Regulator stability caps — prevent oscillation in LDO feedback loop
C105,C108,C110,C112	0.1µF / 0402	Beside each sensor IC	High-freq decoupling at sensor supply pins — keeps noise off sensors
C111	4.7µF / 0402	MAX30102 analogue section	Low-freq supply filter — removes voltage ripple that mimics heartbeat
C22, C23	12pF / 0201 DNM	24MHz crystal Y1	Not mounted — CC2640R2F internal capacitance is sufficient

Resistors — Yageo RC0402 + Vishay CRCW0402 | all 0402 (1.0×0.5mm)

Reference	Value	Engineering Purpose
R1	100kΩ (Yageo)	nRESET pull-up — guarantees chip starts in defined (not-reset) state; without this, a floating reset pin would cause random resets
R15, R16	0Ω jumpers (Yageo)	Zero-ohm jumpers — act as PCB wire links for JTAG_TDO/TDI routing; can be cut to re-route signals without PCB revision
R100, R120	10kΩ (Yageo)	Button pull-ups — DIO_0/DIO_1 held HIGH when buttons not pressed; pressing connects to GND giving a clean LOW signal
R103, R104	10kΩ (Yageo)	I^2C bus pull-ups on SCL (DIO_5) + SDA (DIO_6) — I^2C is open-drain; these resistors pull lines HIGH when idle
R110, R111	100Ω (Vishay)	Series ESD protection on SW101/SW102 — limits current if GPIO pin is accidentally driven hard; protects CC2640R2F
R17, R18	DNM	Reserved configuration pads — unpopulated in Rev1; provides future routing flexibility without PCB respin

Communication Architecture — I²C + BLE + JTAG

Three separate communication channels — each serves a different purpose

I²C Bus — Sensor Communication (DIO_5=SCL, DIO_6=SDA, 400kHz fast mode)

How I²C works: Master sends START + 7-bit slave address + R/W bit → addressed slave responds with ACK → data bytes exchanged → STOP. Only 2 wires for any number of devices.

BLE 5.1 — Wireless Data Transmission to Phone / Gateway

CC2640R2F → RF_P/RF_N → L12(2nH)+L21(15nH)+C14(12pF) impedance matching → PCB trace IFA antenna (TI AN043, 50Ω) → 2.4GHz BLE
Advertises every 100ms (DEFAULT_ADVERTISING_INTERVAL=160 × 0.625ms). Once connected, sends GATT notifications every 350ms with 6-byte sensor data array.
Range: typically 10–30 metres (indoor). Power: ~2mA average current including all sensors.

JTAG — Firmware Programming & Debugging via P1 Header

P1 header (18-pin) → JTAG_TCK(DIO_3) + JTAG_TMS(DIO_4) + TDI + TDO + nRESET → TI XDS110 USB Debug Probe → Code Composer Studio
Used to: flash firmware (.hex/.out files), set breakpoints, inspect memory registers, step through code line by line, and measure power consumption.

Teaching note — What is GATT and why does BLE use it?

GATT (Generic Attribute Profile) organises BLE data into Services and Characteristics — like a menu. This device exposes Service 0xFFF0 with 5 characteristics (CHAR1–CHAR5). CHAR1 holds the 6-byte sensor array. A phone app simply reads CHAR1 every 350ms — no custom protocol needed. Any BLE scanner app (nRF Connect) can read raw sensor data from this band without any custom app.

PCB Layout Principles — Design Decisions Explained

Understanding WHY components are placed where they are — key embedded PCB design knowledge

Sensor placement — bottom side

MAX30102 (HR) and MAX30205 (temperature) are mounted on the PCB bottom (solder side). When worn on the wrist, the bottom faces the skin — giving the optical sensor direct skin contact for PPG measurement and the temperature sensor direct skin contact for body temperature. This is not accidental — it is a fundamental hardware design decision.

Decoupling capacitors — placed beside each IC

Every IC has 0.1µF and/or 10µF capacitors placed as physically close as possible. This is critical: when a digital IC switches states, it draws a brief current spike. Without a nearby capacitor, this spike causes a voltage dip on the power rail. The cap acts as a local charge reservoir — it supplies the spike current instantly. The larger the distance to the cap, the more inductance in the path and the less effective it is.

Separate power planes — VDDS and 3V3

The PCB has separate copper power planes for VDDS (CC2640R2F internal supply) and 3V3 (external supply). This prevents digital switching noise from the MCU coupling onto the 3V3 rail that powers the analogue sensors. The ferrite bead FL1 adds a further barrier between the external supply input and the internal planes — a standard mixed-signal PCB technique.

RF section — keep-out zone and 50Ω line

The PCB trace antenna requires a ground-plane keep-out zone underneath it — a copper-free area so the antenna radiates upward rather than shorting into ground. The transmission line between the RF matching network (L12/L21/C14) and the antenna is carefully sized for 50Ω characteristic impedance. Any mismatch reflects power and reduces range.

4-layer PCB stack-up

The original research paper (Pinheiro et al. 2022) specifies a 4-layer PCB: top copper (components + antenna), inner ground plane (EMI shield), inner power plane (stable reference), bottom copper (sensors). The ground plane between the digital and analogue layers shields sensor circuits from digital switching noise — essential for clinical-grade accuracy.

Component packages — 0201 and 0402

Most passive components use 0201 (0.6×0.3mm) or 0402 (1.0×0.5mm) packages — tiny SMD components requiring reflow soldering. This miniaturisation enables the 34×40mm form factor. RF components (L12, L21, C14) use 0201 specifically — at 2.4GHz even a 1mm trace has significant inductance, so minimising component size is essential for RF performance.

How Everything Works Together

End-to-end data flow — from skin contact to smartphone screen

1

Sensor measures vital sign

MAX30102 shines LEDs through wrist skin and photodetects reflected light. MAX30205 measures skin temperature via bandgap. LIS331DL detects wrist acceleration. Each sensor generates digital data.

2

CC2640R2F reads via I^2C

Firmware (simple_peripheral.c) running on the ARM Cortex-M3 sends I^2C read commands every 350ms. Addresses: 0x57 (HR), 0x90 (temp), 0x19 (accel). Data arrives as raw bytes via DIO_5/DIO_6.

3

Firmware processes data

Raw bytes converted to meaningful values: temp × 0.00390625 = °C. PPG signal processed for HR. Accelerometer data used for step counting and motion artifact removal from PPG.

4

BLE GATT characteristic updated

6-byte array [sensor_id, accel_whoami, accel_xh, accel_xl, temp_high, temp_low] written to CHAR1 (UUID 0xFFF1). TI BLE stack automatically sends GATT notification to connected phone.

5

BLE radio transmits

Data modulated onto 2.4GHz BLE carrier. RF matching network (L12/L21/C14) ensures maximum power transfer to PCB trace IFA antenna. Signal propagates wirelessly up to 30m.

6

Smartphone / gateway receives

Phone app or Raspberry Pi gateway receives GATT notification. Data decoded, displayed in real-time. Healthcare worker monitors patient vitals remotely — no physical contact needed.

Power budget: ~2.0mA average (all sensors + BLE active) | <1µA deep sleep | 150mAh LiPo → ~73 hours continuous operation (Pinheiro et al. 2022)

Key Takeaways

Band01_Rev1 — 52-Component BLE Health Wearable | Project Work Task 1

- CC2640R2F is a 32-bit ARM Cortex-M3 BLE 5.1 MCU — central processor reading sensors via I^2C and transmitting data wirelessly
- 3 sensors on one shared I^2C bus: MAX30102 (PPG heart rate + SpO2 via LED/photodetector), MAX30205 (±0.1°C clinical temperature), LIS331DL (MEMS accelerometer)
- Dual LP5907 ultra-low-noise LDO regulators (3.3V + 1.8V) with firmware-controlled EN pins — enables <1µA deep sleep and 73-hour battery life
- Inverted-F PCB trace antenna (TI AN043) + L12/L21/C14 RF matching network + FL1 ferrite bead — complete BLE 2.4GHz RF subsystem
- 52 components confirmed in Aisler BOM (March 2026) from TI, Analog Devices, STMicro, Murata, NDK, ECS, Samtec, Yageo, Vishay, Samsung, KEMET, Taiyo Yuden, Panasonic
- Next: Task 2 — firmware analysis in Code Composer Studio. Every pin mapping (DIO_5=SCL, DIO_8=interrupt…) and power rail studied here maps directly to firmware code